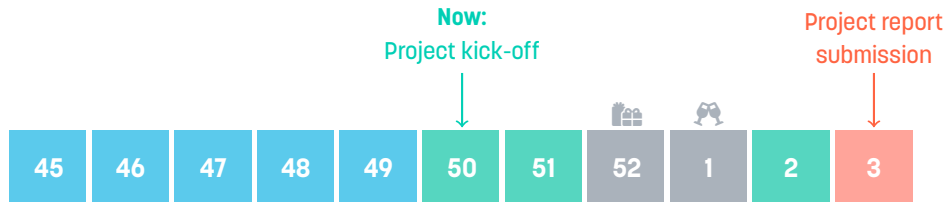


Project kick-off

Marcel Bollmann

Department of Computer and Information Science (IDA)

Project timeline



- Please use the chance to get **individual supervision!**
 - No meetings in week 52 or week 1.
 - There will be some slots for meetings again in weeks 2–3.

Plan for today

1. Formal Requirements

2. Example Projects

3. Practical Tips

- Identify your problem
- Design your approach
- Evaluate your approach
- Produce your report
- Getting help

Formal Requirements



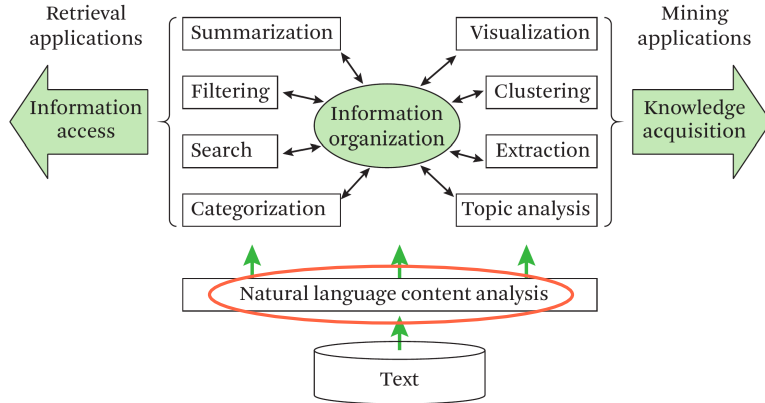
Purpose of the project

The project module primarily tests to what extent you are able to

- 1 **identify, formulate and solve problems** *within the area of text mining*, and
- 2 **clearly present and discuss the conclusions** of a project work.

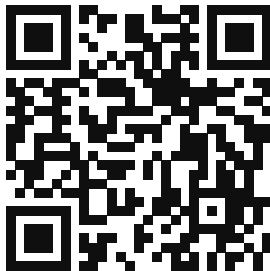
From the course memo

Is your problem “within the area of text mining”?



Zhai and Massung (2016)

Make sure you are familiar with the requirements!



<https://liu-nlp.ai/text-mining/project/>

Submission

- The project report is submitted as a single PDF file **via Lisam**.
- Additionally, I will prepare a **project registration form** for you to submit:
 - Your project title & abstract. *(can be published as example projects in coming years)*
 - A link to your code repository.
 - Either on Github or on LiU's Gitlab server.
 - For private repos, you must give access to mbollmann (Github) or marbo59 (Gitlab).
 - A declaration of whether and how you used generative AI in preparing your report.

Example projects from previous years



Sentiment Analysis of IMDB Movie Reviews

- How well can we **predict the sentiment** of movie reviews?
 - Used the existing *IMDB movie reviews* dataset
- Trained and evaluated different types of classifiers.
 - Naive Bayes, Support Vector Machine, BERT, RoBERTa
- Performed a manual **error analysis** of reviews misclassified by BERT/RoBERTa.
 - Gain additional insights that could help improve models further

Explainable Authorship Attribution using BERT

- Can we **identify an author** from the way they write?
 - Found an *author identification dataset* on Kaggle
- Trained and evaluated different types of classifiers.
 - Naive Bayes, Logistic Regression, BERT
- Which features does the BERT model consider “**characteristic**” for each author?
 - Used *integrated gradients method* to extract features that were most important for the BERT classifier when making its predictions

Analyzing how the Refugee Crisis Affected Parliament Speeches using BERTopic

- Did the Syrian refugee crisis cause an increase in **immigration-related topics** being discussed in the Swedish parliament?
- Analysed the change of topics over time, as modelled by **BERTopic**.
 - BERT-based *topic modelling* technique
- Evaluated the performance using a **coherence** metric as well as manual analysis.

Generating free-text explanations with GPT-2/GPT-3

- Natural language inference (NLI) is the task of predicting whether two sentences **entail or contradict** each other.
 - “*An adult holds a stick.*” contradicts “*An adult walks away empty-handed.*”
- Can we **generate free-text explanations** for these predictions?
 - “*Holding a stick implies using hands, so it is not empty-handed.*”
- Compared **two generative models** using different techniques and performed **human evaluation** on the generations.
 - GPT-2 with fine-tuning vs. GPT-3 with few-shot prompting

Family tree extraction for Tolkien's world

- Can we automatically **extract family trees** for Tolkien's characters using encyclopedia entries about them?
 - Scraped the *Lord of the Rings Wikia* site
 - Implies **detecting** character names and the **relations** between them
- Evaluated using the **“infoboxes”** section of each character page as ground truth.

Project abstracts from previous years

- A selection of **project abstracts** from previous years is on the course website.

Note

It is perfectly fine to work on the same topic as a project from previous years!

- Perhaps you can find a better approach, or do a different analysis, or...
- You are unlikely to have *exactly* the same ideas as another student.

Practical Tips



A common project structure

- 1 Identify your problem (ca. 8 hours)
- 2 Design your approach (ca. 32 hours)
- 3 Evaluate your approach (ca. 32 hours)
- 4 Produce your report (ca. 16 hours)

Phase 1: Identify your problem

- **Task-focused:** Is there a specific task or method you want to work on?
 - *“relation extraction to construct family trees of Tolkien’s characters”*
 - This kind of project might ask “which model performs best on task [X]?”
- **Question-focused:** Is there a small research question you want to answer?
 - *“Did the Syrian refugee crisis cause an increase in immigration-related topics?”*
 - Focus on what we can learn from the data; often more exploratory in nature!
- One way to get inspiration is to **look for datasets** first.
 - [↗ Kaggle](#), [↗ HuggingFace Datasets](#), [↗ Riksdagens öppna data](#), many “shared tasks” such as [↗ RepEval 2017](#), ...

Phase 2: Design your approach

- 1 Select one or more **datasets**.
- 2 Decide on a **method** and prepare code to run your models/experiments.
 - Use existing libraries, e.g. [spaCy](#), [Gensim](#), [HuggingFace Transformers](#), ...
 - No requirements on the programming language or libraries that you can use!
- 3 Choose an approach for **evaluating** and analysing your results.
 - Consider both evaluation *measures* as well as *baselines*.
 - Think about this **before** implementing your approach!

Phase 2: Design your approach – tips & tricks (I)

- It's a good idea to **review previous work** related to your problem.
 - Search the [ACL Anthology](#) (*many NLP & text mining papers*)
 - Search via [Semantic Scholar](#)
 - **New:** Search via [Scopus AI](#) (*licensed through LiU's library*)
- This can give you new ideas as well as pointers to existing **code and datasets!**
- Collect references as you go so you can **cite your sources** appropriately!
 - Including code that you re-used

Phase 2: Design your approach – tips & tricks (II)

If your project uses **machine learning** in any way, consider:

-  **How to avoid machine learning pitfalls, by Michael A. Lones**
 - Before you start to build models
 - How to reliably build models
 - How to robustly evaluate models
 - How to compare models fairly
 - How to report your results

Phase 2: Design your approach – tips & tricks (III)

- **Web scraping** is the process of automatically extracting data from websites.
- Involves fetching a web page, parsing it, and finally extracting data from it.
 -  **BeautifulSoup**
 -  **Scrapy**
 -  **Trafilatura**
- May violate the terms of use of some websites and/or constitute copyright infringement.
 - Countermeasure include blocking the scraper's IP address.

Phase 2: Design your approach – common questions

- 1 Can we use methods or **topics from outside the course**?
 - **Absolutely**, as long as it involves the analysis of natural language content.
- 2 Does the project have to have **scientific novelty**?
 - **No!** You don't need to produce novel results.
- 3 Are there **computing resources** we can use?
 - We don't have dedicated computing resources for student projects. 😞
 - Consider free resources like [Colab](#) or [Kaggle](#).
 - You might try lab rooms *Asgård*, *Egypten*, *Olympen* outside of course bookings.

Phase 3: Evaluate your approach

- Run your experiments and **interpret your results**.
 - **Quantitative** measures, *e.g.*, accuracy, precision, recall, topic coherence, perplexity, ...
 - **Qualitative** analysis, *e.g.*, human evaluation, ablation studies, manual inspection of interesting test cases, ...
 - This can be an absolutely valid way of improving your analysis!
- Remember that most evaluation methods are **relative**!
 - Can't conclude that "X works well", but "X works *better* or *worse* than Y"

Phase 4: Produce your report

- Use the **required template!**
- Do **not** go over the **page limit!**
 - 3–5 pages
 - References don't count.
 - Appendix doesn't count, but your paper must be complete without it.

Instructions for *ACL Proceedings		
Anonymous ACL submission		
001	Abstract	To load the style file in the review version:
002	This document is a supplement to the general	<code>\usepackage[review]{acl}</code>
003	instructions for *ACL authors. It contains in-	007
004	structions for using the L ^A T _E X style files for	For the final version, omit the review option:
005	ACL conferences. The document itself con-	<code>\usepackage{acl}</code>
006	forms to its own specifications, and is there-	009
007	fore an example of what your manuscript should	To use Times Roman, put the following in the
008	look like. These instructions should be used	preamble:
009	both for papers submitted for review and for	<code>\usepackage{times}</code>
010	final versions of accepted papers.	043
011	1 Introduction	(Alternatives like txfonts or newtx are also accept-
012	These instructions are for authors submitting pa-	able.)
013	pers to *ACL conferences using L ^A T _E X. They are	Please see the L ^A T _E X source of this document for
014	not self-contained. All authors must follow the gen-	comments on other packages that may be useful.
015	eral instructions for *ACL proceedings, ¹ and this	Set the title and author using <code>\title</code> and
016	document contains additional instructions for the	<code>\author</code> . Within the author list, format multiple
017	L ^A T _E X style files.	authors using <code>\and</code> and <code>\and</code> and <code>\AND</code> ; please see
018	The templates include the L ^A T _E X source of this	the L ^A T _E X source for examples.
019	document (<code>acl_latex.tex</code>), the L ^A T _E X style file	By default, the box containing the title and au-
020	used to format it (<code>acl.sty</code>), an ACL bibliography	thor names is set to the minimum of 5 cm. If
021	style (<code>acl_natbib.bst</code>), an example bibliography	you need more space, include the following in the
022	(<code>custom.bib</code>), and the bibliography for the ACL	preamble:
023	Anthology (<code>anthology.bib</code>).	<code>\setlength\titlesbox{<dim>}</code>
024	2 Engines	005
025	To produce a PDF file, pdfL ^A T _E X is strongly recom-	where <code><dim></code> is replaced with a length. Do not set
026	mended (over original L ^A T _E X plus <code>dvips</code> or <code>pdflatex</code>).	this length smaller than 5 cm.
027	The style file <code>acl.sty</code> can also be	007
028	used with <code>lualatex</code> and <code>X_{La}TeX</code> , which are es-	4 Document Body
029	pecially suitable for text in non-Latin scripts.	008
030	The file <code>acl_luatex.tex</code> in this repository provides	4.1 Footnotes
031	an example of how to use <code>acl.sty</code> with either	Footnotes are inserted with the <code>\footnote</code> com-
032	<code>lualatex</code> or <code>X_{La}TeX</code> .	mand. ²
033	3 Preamble	009
034	The first line of the file must be	4.2 Tables and figures
035	<code>\documentclass[11pt]{article}</code>	See Table 1 for an example of a table and its caption.
	¹ http://acl-org.github.io/ACL2018/formatting.html	Do not override the default caption sizes.
	044	As much as possible, fonts in figures should
		conform to the document fonts. See Figure 1 for
		an example of a figure and its caption.
		007
		² This is a footnote.

Phase 4: Produce your report – tips & tricks (I)

- You can use the **suggested structure**, but it's absolutely not required!
 - The “ideal” structure can vary depending on what your project is about.
- **Don't** explain commonly known methods in **too much detail**.
 - Reproducing *e.g.* formulas for precision/recall, Naive Bayes, etc. *does not add any value*.
 - Your description should be complete enough to understand what you did.
- Do think about **how to present** your project in an understandable way.
 - *e.g.* figure with data examples (inputs/outputs), figure illustrating a non-trivial concept
 - *e.g.* visualizing interesting results in a graph or plot

Phase 4: Produce your report – tips & tricks (II)

Phase 4: Produce your report – common questions

- 1 What is the expected **amount of sources** to cite?
 - **There is none**, but you must cite all sources of text/data/code/ideas that you used.
 - For a higher grade, you need to cite scientific articles, but **quality is more important** than quantity. Citing one paper with an insightful and meaningful discussion is much better than just citing ten without any meaningful connection to your work.

- 2 My project is really complex, can I go **over the page limit**?
 - **No!** We may return your report without grading.
 - If it is not possible to fully describe & present your project in max. 5 pages, it is probably too big for this course! → Talk to me early if unsure about the scope.

How to get help

- Pitch your project idea to me!
- I am offering **one-on-one meetings** for you to get feedback.
 - Book a time slot through the link on the website
- I am also available **via e-mail**.
 - Use whichever option works best for you and the question(s) you have!

